

RRDS26 Workshop 2 Activity 3

Personal Quarto Website

Setup

1. We will begin by setting up a Quarto website project or folder. Instructions are below and can also be found on this page: <https://quarto.org/docs/websites/>.

In RStudio, do the following to create a Quarto Website Project:

- a. Click on the Project option in the top right of the program.
- b. Select New Project....
- c. Select New Directory.
- d. Select Quarto Website.
- e. Put the directory name as Website (or something similar) and create it as a subdirectory of where you want it to be located, like an RRDS26 folder (it is best to not put it inside the RRDS26-Workshop2 folder you created in Activity 2). You can keep the default options and select Create Project.

This will create a folder with `_quarto.yml`, `about.qmd`, `index.qmd`, `.Rproj`, and `styles.css` files.

In Positron, do the following to create a Quarto Website folder:

- a. Open the Command Palette by typing Command+Shift+P on a Mac or Control+Shift+P on Windows.
- b. In the Command Palette, type Quarto: Create Project and select that option.
- c. Click on Website Project.
- d. Navigate to the folder you want to host your website folder on your computer, like an RRDS26 folder (it is best to not put it inside the RRDS26-Workshop2 folder you created in Activity 2) and click on Choose Parent Directory.
- e. Name the directory something like Website that will store your website files.

This will create a folder with `_quarto.yml`, `about.qmd`, `index.qmd`, and `styles.css` files.

Building Website

2. Now it's time to start building the website.
 - a. Make sure you are in your website project in RStudio or the website folder in Positron. You will know because the name of project (or folder in Positron) will appear in the top-right of the window.
 - b. Open the `_quarto.yml` file (it might already be open by default) in your IDE.
 - c. The first two lines should currently be

```
project:
  type: website
```

Modify this so the first three lines are

```
project:
  type: website
  output-dir: docs
```

Note that the `type` and `output-dir` lines are both indented.

- d. Now, you can add Quarto files as pages to your website. Under the
 - **about.qmd** line in the **_quarto.yml** file, you can add lines to add pages to the site. For example, adding
 - **href: Activities/Activity 1.qmd**
 - text: Activity 1**
- will add a page with your first activity from this workshop on it. Of course, make sure to actually add the Activities folder and Activity 1.qmd file in your Website directory and make sure the name of the .qmd file matches the one that is actually in the Activities folder.

Deploying Website

3. Now we need to render the website.

In RStudio:

- a. When working in the website project environment, there should be a Build pane near the Environment pane near the top right of the window. Click that.
- b. Click Render Website. This will render all of the Quarto files, so it could take a little while if any one has a lot of code. If it takes too long, or gives problems when rendering it, you could put the .html files instead of .qmd files in Step 2d above, but it doesn't work as well as using the .qmd file since it will lack the website metadata like a navigation bar.
- c. Once it renders, this should give you a preview of your website in the Viewer pane. You can click the Show in new window button (with an arrow pointing up and right) to open it in a new window so you can navigate the website. This should also add a docs/ folder with files in it among other things.

In Positron:

- a. When working in the website project folder, open the command palette (Command + Shift + P or Control + Shift + P) and type Quarto: Render Project. This will render all of the Quarto files, so it could take a little while if any one has a lot of code. If it takes too long, or gives problems when rendering it, you could put the .html files instead of .qmd files in Step 2d above, but it doesn't work as well as using the .qmd file since it will lack the website metadata like a navigation bar.
- b. You can view the website in Positron's viewer by opening the index.qmd file and clicking Preview or typing Command + Shift + K (Mac) or Control + Shift + K (Windows).

4. Now we want to link the website folder with GitHub:

- a. In GitHub Desktop, click on the Current Repository button, click Add, and then Create New Repository....
- b. Give it the same Name as the website folder you created in part 1 above and make sure the Local Path leads to its parent directory. You can add a README if you'd like, but don't add a .gitignore yet. You can add one later, but there are files we don't want ignored in our website directory that would be ignored if we add the .gitignore now.
- c. Now click on the Publish repository button. Give it the name you want for the repo on GitHub, give a description if you'd like (optional), and uncheck the Keep this code private option. GitHub pages does not work with private repos. The URL for the website will be

- GitHubUsername.github.io/WebsiteRepoName unless you specifically name the repo GitHubUsername.github.io. In that case, the URL will be GitHubUsername.github.io.
- d. Click Publish Repository and it will publish on GitHub.
5. On GitHub, go to your website directory and do the following:
- a. Click on Settings.
 - b. Go down to Pages on the left side.
 - c. Under Branch, select None and change it to main.
 - d. Change /(root) to /docs.
 - e. Click Save.
6. After a moment or two, the page should deploy. When back on your main website repo page, on the right side it should have a **Deployments** section with github-pages underneath. Click on the github-pages link.
7. Then click on the URL under git-hub pages to see your website.

Now when you make changes to the files and push them to GitHub, it will automatically deploy the updated website (it may take a few minutes after each push to update it).

There is so much customization you can do with your website! You can check out one I made here: <https://rbrown53.github.io>. The repository for it with the **_quarto.yml** file can be found here: <https://github.com/rbrown53/rbrown53.github.io>. Feel free to look at that file and experiment with your website. There is also code on there to include pdf files and shiny apps if you have any you need to include.

There is also lots of helpful info on the Quarto help pages: <https://quarto.org/docs/websites/website-navigation.html>.

Closing Project or Folder

8. **In RStudio:**

When you are finished, you can close the project by clicking on the project name in the top right of the window and going down to Close Project. You can reopen the website project by opening the .Rproj file in your website folder. If you don't close this, it may automatically open the next time you want to use RStudio, which you likely don't want.

In Positron:

You can close the folder by clicking on the folder name in the top right of the window and going down to Close Folder. You can reopen the website folder by clicking that folder icon in the top right again and navigating to your website folder. If you don't close this, it may automatically open the next time you want to use Positron, which you likely don't want.